

2024 Bi H1 Q19

Section: Sustainability and Interdependence

Topic: Photosynthesis

Question Summary:

The question asks for the role of NADPH in the Calvin cycle of photosynthesis.

Worked Solution:

- The Calvin cycle takes place in the stroma of the chloroplast.
- ATP provides energy, while NADPH provides hydrogen and electrons for reduction reactions.
- NADPH reduces 3-phosphoglycerate (3PG) to glyceraldehyde-3-phosphate (G3P) by donating hydrogen and electrons.

Therefore, NADPH's role in the Calvin cycle is to add hydrogen to 3-phosphoglycerate (3PG).

Final Answer: **C** (Add hydrogen to 3-phosphoglycerate).

Revision Tips:

- Calvin cycle uses ATP for energy and NADPH for reduction.
- 3PG → G3P → glucose or starch.
- Light reactions produce ATP and NADPH; dark reactions use them.